

4. PLACEORDER SPECIFICATION

4.1 Purpose

placeOrder creates a new order for a user using the selected food items and requested quantities. It coordinates with the User, Catalogue, and Payment services while hiding its internal implementation details from the caller.

4.2 Inputs and Outputs

Inputs	Outputs
• User reference • Selected food items • Quantity for each selected item • Delivery details • Payment method	• Order confirmation • Order details • Final order amount • Payment result

4.3 Error Cases

- User not found
- Invalid food item
- Food item unavailable
- Invalid quantity
- Invalid delivery details
- Payment failed
- Order could not be created

4.4 Internal Details Hidden from Callers

The caller does not need to know:

- How the order is stored internally
- How order items are stored
- How food availability is checked internally
- How the final amount is calculated internally
- How payment processing is implemented
- Which database tables are used
- Which SQL queries are executed
- How services communicate internally

The caller only interacts with the **placeOrder** contract and receives the defined output or an appropriate error.

4.5 Operation Flow

Step	Action	Handled by
1	Receive the placeOrder request.	Order Service
2	Validate the user.	User Service — getUser
3	Check the selected food items.	Catalogue Service — checkItems
4	Calculate the order amount.	Order Service
5	Request payment.	Payment Service — createPayment
6	Create the order if payment succeeds.	Order Service
7	Return the order confirmation to the caller.	Order Service